MINIMAL ORIENTATIONS IN A SEVEN-CYCLE FAMILY OF CRITICAL GRAPHS

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Abstract. Hanson and Toft observed that a seven-cycle construction gives a k-critical graph for each $k \geq 4$. The first two graphs were known to have no orientation with exactly one directed path of length $k-1$; Jensen and Toft asked whether this holds throughout the family. We prove that it does. A short identity for color classes on the cycle reduces the infinite statement to the six cases $4 \leq k \leq 9$. For these cases we give Boolean encodings of a unique full increasing path, unsatisfiability certificates, and a small independent checker that reconstructs each formula and verifies its proof.

1. Introduction

An orientation of a k-chromatic graph is *minimal* if it has exactly one directed simple path of length $k-1$. Gallai asked whether every critical graph has a minimal orientation [1]. Youngs answered this question negatively, beginning with a 14-vertex 4-critical graph [3]. Hanson and Toft identified this graph as the first of a seven-cycle family H_k. Jensen and Toft recorded the negative cases $k = 4, 5$ and asked whether the entire family has no minimal orientation [2, §13.8, pp. 215–216]. We answer that family question.

Our proof has two parts. An elementary identity concerning the seven attachment triples permits a common color to be deleted from every clique whenever $k \geq 10$. Every full increasing path of the smaller graph then extends injectively to one of the larger graph. The cases $k = 4, \ldots, 9$ are finite: their exact unsatisfiable formulas and checkable proof traces are included in the accompanying supplementary archive. The reduction, formula semantics, and certificate format are specified below so that the finite computation is a verifiable premise of the theorem.

Write $C = x_0x_1 \cdots x_6x_0$, with indices in $\mathbb{Z}/7\mathbb{Z}$. For each i, take a clique Q_i of order $k-3$, with the seven cliques mutually disjoint and disjoint from C. Join every vertex of Q_i to

$$T_i = \{x_i, x_{i+1}, x_{i+3}\}, \tag{1}$$

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and add no other edges. Denote the resulting graph by H_k. Direct counting gives

$$(2)\quad |V(H_k)| = 7(k-2), \qquad |E(H_k)| = \frac{7}{2}(k^2-k-4), \qquad \Delta(H_k) = 3k-7.$$

An *ordered k-coloring* is a proper vertex coloring by the ordered palette $0,\dots,k-1$. A *full increasing path* is a simple path whose vertices have colors $0,1,\dots,k-1$ in that order. We use the following orientation–coloring criterion stated in [2, §13.8]: a k-critical graph admits a minimal orientation if and only if it has an ordered k-coloring with exactly one full increasing path. The forward implication applies to *arbitrary* orientations: Gallai's reduction first gives an acyclic minimal orientation and then its longest-path levels give the coloring. Conversely, orienting each edge from its smaller to its larger color gives the desired acyclic orientation. Thus it suffices to prove that every ordered k-coloring of H_k has at least two full increasing paths.

2. The cycle identity and criticality

For a proper coloring of C, let q be its number of distinct colors and let t be the number of triples T_i that use exactly two colors. Each triple uses either two or three colors because it contains the cycle edge x_ix_{i+1}.

Lemma 1. *For every proper coloring of C,*

$$(3)\qquad t \le 6 \qquad \text{and} \qquad q+t \le 9.$$

Proof. The nonedge pairs in T_i are $\{x_i, x_{i+3}\}$ and $\{x_{i+1}, x_{i+3}\}$. As i varies, these are exactly the fourteen nonedges of C, each occurring once: the first type has cyclic distance three and the second distance two. Since a triple contains an edge, it is two-colored precisely when one of its nonedge pairs is monochromatic. Its other nonedge pair cannot also be monochromatic. Consequently, if $\mathcal{C}$ is the set of cycle color classes, then

$$t = \sum_{A\in\mathcal{C}} \binom{|A|}{2}.$$

Every color class is independent and has at most three vertices. If a_j counts classes of size j, then

$$7 = a_1 + 2a_2 + 3a_3, \qquad q = a_1 + a_2 + a_3, \qquad t = a_2 + 3a_3.$$

Hence $q + t = 7 + a_3 \le 9$, since $a_3 \le 2$. Also $t \le 6$: when $a_3 = 0, 1, 2$, respectively, the above first equality bounds t by $3, 5, 6$. □

Proposition 2. *For every $k \ge 4$, the graph H_k is fully k-critical: it has chromatic number k, and every proper subgraph is $(k-1)$-colorable.*

Proof. In a $(k-1)$-coloring of H_k, each Q_i would need $k-3$ different colors outside the colors on T_i. Thus all seven triples would have at most two colors, contradicting $t \le 6$. Conversely, three-color the cycle and give each

Q_i any $k-3$ colors absent from T_i out of a k-color palette. This proves $\chi(H_k) = k$.

For criticality, color each edge deletion using $k-1$ colors. Upon deleting a cycle edge, two-color the resulting seven-vertex path alternately. All triples then use at most two colors, leaving at least $k-3$ colors for each clique.

For the other edges, the proper cycle coloring

$$(\mathrm{col}(x_0), \ldots, \mathrm{col}(x_6)) = (0, 1, 0, 2, 1, 0, 1) \tag{4}$$

has T_0 as its only three-color triple. Rotate this pattern so that T_i is the exceptional triple. Every other clique can use $k-3$ colors absent from its two-color triple. If the deleted edge joins $y \in Q_i$ to $x_j \in T_i$, give y the color of x_j, and give the remaining $k-4$ vertices of Q_i distinct colors outside the three colors of T_i. If the deleted edge is uv inside Q_i, give u, v the same color outside T_i and give the other $k-5$ vertices distinct remaining colors outside T_i. This last case occurs only when $k \geq 5$, so such a color exists. In each case the stated coloring is proper after deleting the specified edge.

There are no isolated vertices. For a vertex v, restrict a $(k-1)$-coloring after deletion of any incident edge to $H_k - v$. Every proper subgraph is contained in an edge or vertex deletion, so the claim follows. □

3. A reduction to six cases

Lemma 3. *Suppose an ordered k-coloring of H_k has a color c absent from C and present in every Q_i. Delete the seven vertices of color c and remove that rank from the palette. This gives an ordered $(k-1)$-coloring of H_{k-1}. Every full increasing path of the smaller coloring extends, injectively, to a full increasing path of the original coloring.*

Proof. Each Q_i is a clique, so it has exactly one vertex of color c. Their deletion reduces all seven clique orders from $k-3$ to $k-4$; the cycle and every attachment relation remain unchanged.

Every edge of the smaller graph has both ends in a block $B_i = Q_i \cup T_i$. For an edge in or incident with a clique this is immediate. The cycle edge $x_i x_{i+1}$ belongs to B_i. Further, every vertex belongs to at least one block.

Take a full increasing path P of the smaller coloring. If c is an interior rank, the vertices immediately below and above c on P are consecutive and therefore adjacent. Choose a block B_i containing this edge. Its deleted color-c clique vertex is adjacent to both ends of the edge; insert it between them. If c is an extreme rank, choose a block containing the first or last vertex of P, respectively, and insert its deleted vertex at that end. The result is a simple full increasing path in H_k. Fix the least eligible block index to make the extension a function. Removing the unique color-c vertex recovers P, so distinct paths have distinct extensions. □

Lemma 4. *Every ordered k-coloring of H_k, $k \geq 10$, has a color absent from C and present in every Q_i.*

Proof. Let q, t be as in Lemma 1. There are $k-q$ colors absent from C. A three-color T_i forces Q_i to use all $k-3$ colors outside that triple. A two-color T_i allows $k-2$ colors and its clique omits exactly one of them. Thus at most t colors absent from C can be omitted by at least one clique. By (3), $k-q>t$ when $k \geq 10$, proving the claim. □

Every ordered h-coloring of an h-chromatic graph has at least one full increasing path. Indeed, orient its edges toward larger colors, and assign to each vertex the number of vertices in a longest directed path ending there. These numbers properly color the graph. If no path had h vertices, they would give an $(h-1)$-coloring, a contradiction. In the color orientation a path has at most h vertices, and a path with h vertices uses every rank.

It follows from Lemmas 3 and 4 that if some coloring of H_k, $k \geq 10$, had exactly one full increasing path, then its restricted coloring of H_{k-1} would have at most one. By Proposition 2 and the preceding paragraph it would have at least one, hence exactly one. Repeating this argument reaches H_9. We now rule out such a coloring for every H_k with $4 \leq k \leq 9$.

4. Finite certificates

For reproducibility, label x_i by i, $0 \leq i < 7$, and label the vertices of Q_i consecutively from $7+i(k-3)$ to $7+(i+1)(k-3)-1$. For each $k = 4, \ldots, 9$, the accompanying CNF asks whether H_k has an ordered coloring with exactly one full increasing path. Its graph edges are generated from (1). We describe both the path encoding and every symmetry restriction used.

Let $C(v,j)$ mean $\mathrm{col}(v) = j$, where $0 \leq j < k$. The formula gives each vertex exactly one color and makes adjacent vertices differently colored. For each vertex v and rank j, let $R(v,j)$ assert that at least one path colored $0, \ldots, j$ ends at v, and let $M(v,j)$ assert that at least two distinct such paths end at v. At the first rank, the exact semantics are

$$R(v,0) \longleftrightarrow C(v,0), \qquad \neg M(v,0).$$

For $j > 0$, writing $N(v)$ for the neighbors of v, impose

$$R(v,j) \longleftrightarrow C(v,j) \wedge \bigvee_{u \in N(v)} R(u,j-1), \tag{5}$$

$$M(v,j) \longleftrightarrow C(v,j) \wedge \left[\begin{array}{l} \displaystyle\bigvee_{u \in N(v)} M(u,j-1) \\ \vee \displaystyle\bigvee_{\substack{u,w \in N(v) \\ u<w}} \big(R(u,j-1) \wedge R(w,j-1)\big) \end{array} \right]. \tag{6}$$

Paths entering v from different neighbors are distinct, and strictly increasing colors prohibit a repeated vertex. Therefore these recurrences exactly count the relevant path prefixes up to two, including multiple prefixes from one neighbor. Every gate in the CNF is defined in both directions. Finally, the

formula requires exactly one $R(v, k-1)$ and no $M(v, k-1)$. A sequential at-most-one counter enforces the first condition. These constraints express exactly one full path.

The following restrictions preserve completeness. Vertices within a Q_i are true twins, so a clique automorphism sorts their colors in increasing numeric vertex order. Rotation of the cycle and attached cliques permits x_0 to be a minimum-color cycle vertex. Finally one may require

$$\operatorname{col}(x_0) \le \left\lfloor \frac{k-1}{2} \right\rfloor. \tag{7}$$

To see this, let a and b be the minimum and maximum colors on the cycle. Either a or $k-1-b$ is at most the right side of (7). If necessary reverse the entire palette order: reversal bijects full increasing paths with their reversed paths. Then rotate a cycle minimum to x_0 and sort the cliques again. Arbitrary permutations of color ranks are not used, since they need not preserve increasing paths.

Proposition 5 (Certified finite cases)**.** *Every ordered k-coloring of H_k has at least two full increasing paths for $4 \le k \le 9$.*

Proof. The formulas just specified are unsatisfiable. Table 1 reports their sizes and the number of checked lemma additions in the positive-hint LRAT refutations. The supplementary archive contains the exact CNFs, proof files, and a standard-library Python verifier. It rebuilds the graph and each CNF independently of the discovery encoder, compares the clauses in exact order, and checks every LRAT addition by unit propagation using its listed positive hints. Negative RAT hints are rejected. Deletion records can be ignored: retained clauses are independently proved consequences of the original formula. In each of the six cases the checked proof derives the empty clause.

TABLE 1. Finite formulas and independently checked refutations.

k	vertices	edges	variables	clauses	LRAT additions
4	14	28	622	2,274	360
5	21	56	1,791	7,205	1,734
6	28	91	4,066	16,972	4,003
7	35	133	7,951	33,738	6,913
8	42	182	14,034	60,002	11,699
9	49	238	22,987	98,599	16,194

By the exact encoding and the sound symmetry restrictions, a coloring with exactly one full increasing path would satisfy the corresponding CNF, contrary to its refutation. A full path exists by the argument before this section, so every coloring has at least two. □

As a separate finite check, a direct path-counting program enumerates all 41,664 colorings of H_4 up to permutations inside the cliques and 25,452,840 *irreducible* colorings of $H_5, \dots, H_9$, again up to clique permutations. Here irreducible means that no cycle-absent color occurs in every Q_i; by Lemma 3, every omitted coloring inherits two paths from the preceding case. The program found no coloring with fewer than two full paths. This check uses neither the CNF encoder nor the LRAT traces.

Theorem 6. *For every integer $k \geq 4$, the graph H_k is fully k-critical and has no minimal orientation.*

Proof. Criticality is Proposition 2. Proposition 5 gives at least two full increasing paths in every ordered coloring for $4 \leq k \leq 9$. Lemmas 3 and 4, together with the existence of at least one full path in every ordered coloring, propagate this assertion to all $k \geq 10$. The orientation–coloring criterion in the introduction excludes a minimal orientation. □

The result concerns the specific family recorded in [2, §13.8]; Youngs's earlier work already disproved the general assertion that all critical graphs have minimal orientations [3]. Equation (2) also gives the upper bound $f(k) \leq 3k - 8$ for the degree threshold $f(k)$ defined in [2, §13.8]. In particular, the present graphs yield $f(6) \leq 10$ and $f(7) \leq 13$.

Data availability

The supplementary archive supplied with this manuscript contains the six graph edge lists, exact CNFs, positive-hint LRAT proofs, Python source for formula reconstruction and proof checking, and a separate direct enumeration program. From the package root, the single command `python certificates/verify_family.py` checks the cycle and criticality premises, reconstructs all six formulas, and checks all six LRAT refutations. This command requires only the Python standard library and no SAT solver. A public repository location for the archive must be added upon deposit before submission.

References

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